Relational Foundation Models (KumoRFM)

Data AI Lab

Taewook Ham

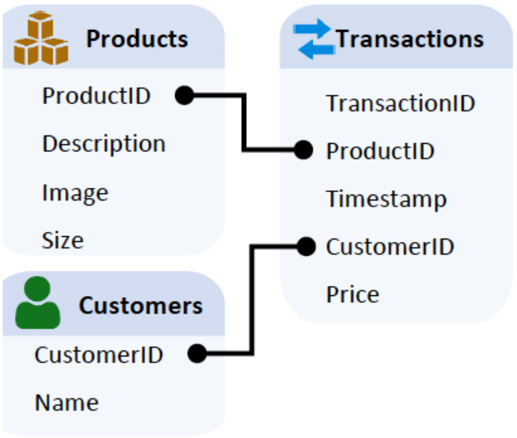
Contents

- **Introduction to Relational Database**
- Relational Deep Learning
- Relational Graph Transformer
- Relational Foundation Model
- Benchmark for RDL
- Toy example

Relational Data(base)

- Most real-world enterprise data lives in **relational database**, not in graphs or text.
  - E-commerce (users, products, transactions)
  - Finance (accounts, transactions, merchants)
  - Healthcare (patients, doctors, treatments)

Histroy in Multiple Table



Basic Terminologies

- Tables
  - A row can be treated as **entity**.
  - A column is an **attribute** of a relational table.
- Keys
  - **Primary key**
  - **Foreign key**
- Relations
  - Arrows as Primary-Foreign Key
- Timestamps
  - Recorded when an entity or event occurred.

Purchase table

Transaction ID	Customer ID	Product ID	Purchange date
1112	24221	8977	03-22-2010
1113	24222	8978	03-22-2010
1114	24223	8979	03-22-2010

Customer ID	Customer	Address
24221	Bob	123 East street
24222	Alice	223 Main street
24223	Martha	465 North street

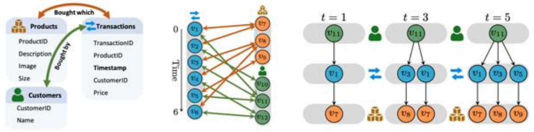
Customer table

Product ID	Name	Price
8977	Banana	.79
8978	TV	400
8979	Watch	50

Product table

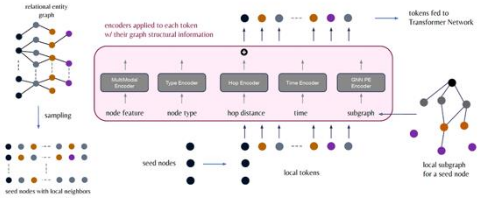
Recent History of Relation Deep Learning

Relational Deep Learning
(2023(ICML 2024))



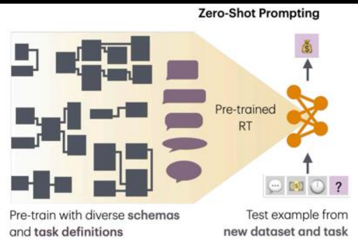
RELBench
2024(NeurIPS2024)

Relational Graph Transformer
(2025(ICLR 2026))



KumoRFM
(2025(From Kumo))

Relational Transformer
(2025(ICLR 2026))



KumoRFM-2
(2026)

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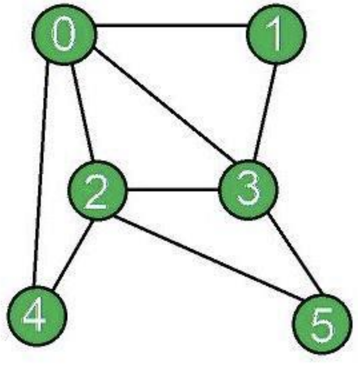
Graphs

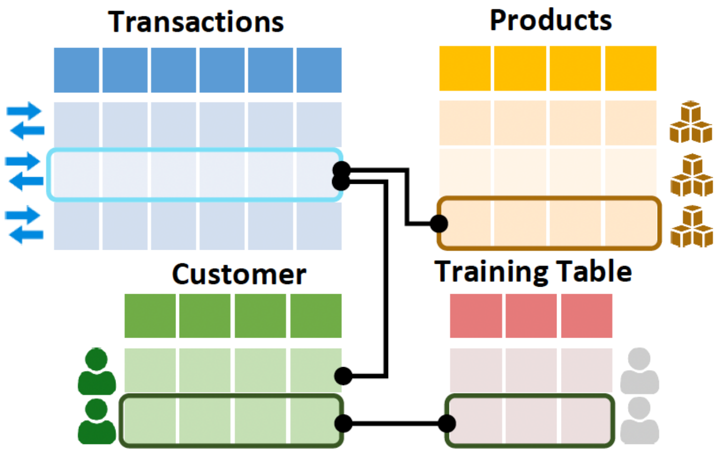
- Data structure that represents connections and relationships.
  - A graph consists of nodes (e.g., users) and edges (e.g., friendship).
  - A graph is represented as a sparse adjacency matrix.
    - $a_{ij} = 1$ if nodes i and j are connected; $a_{ij} = 0$ otherwise.



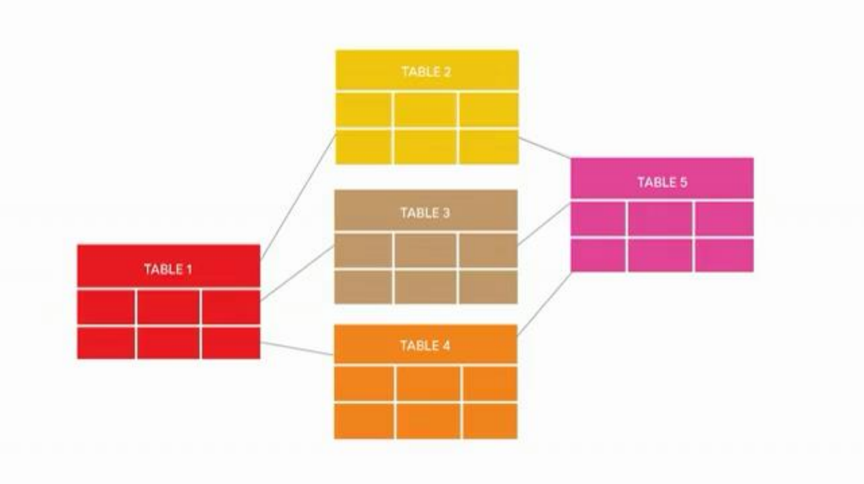
	0	1	2	3	4	5
0	0	1	1	1	1	0
1	1	0	0	1	0	0
2	1	0	0	1	1	1
3	1	1	1	0	0	1
4	1	0	1	0	0	0
5	0	0	1	1	0	0

Relational Entity Graph

- The key is to represent relational database as **temporal, heterogeneous graphs**.
- **Schema graph** defines how tables are related (table-level structure).
- **Relational entity graph** instantiates this schema at the entity (row) level.
  - Each row becomes a node; PK-FK links become edges.



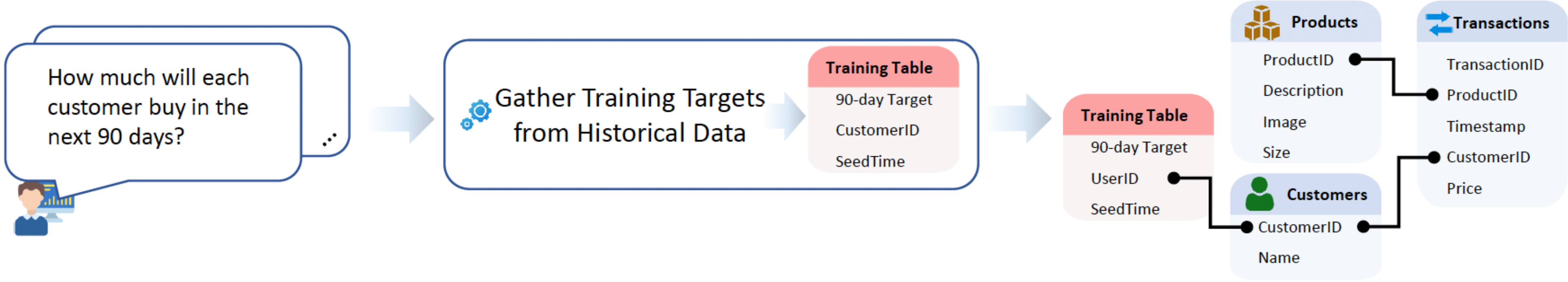
Schema Graph



Relational Entity Graph

Training Table

- Key insight for model training is generating **training table** T_{train}.
  - A training table contains supervision labels **automatically computed** based on historical data in the relational database.
  - Then, T_{train} is linked to the main relational database.



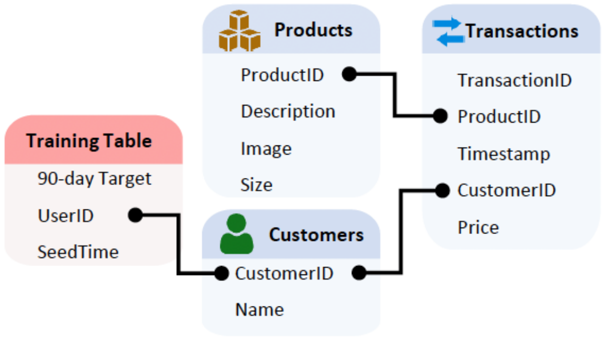
Define Tasks

Training Table Generation

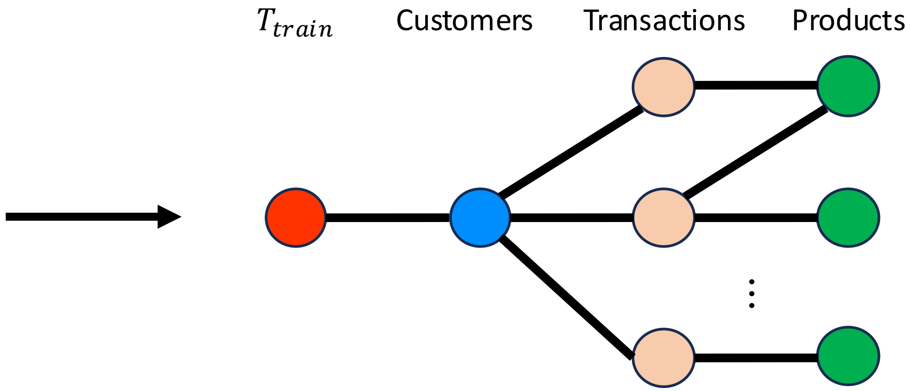
Link to Relational Tables

Time-Consistent Computational Graphs

- **A subgraph is created** induced by the set of foreign keys K_v and its timestamp t_v of a training example in the training table T_{train}.
  - The subgraph acts as a **local** and **time-consistent** computation graph to predict ground-truth label y_v.



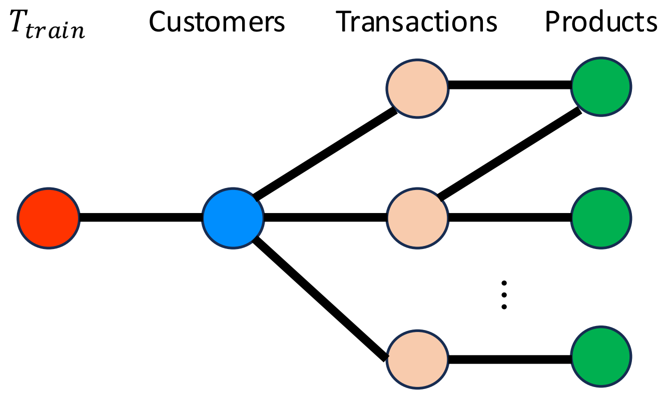
Relational Tables



Subgraph Sampling of a Training Example

Temporal, Heterogeneous Message Passing

- Information flow between nodes **can only go forward in time**, ensured by a temporal neighbor sampler.
- Node embeddings are constructed by encoding **modality-specific column features** and **fusing** them into a unified row-level representation.



Message-Passing Graph Neural Networks (MP-GNNs) (Gilmer et al., 2017; Fey & Lenssen, 2019) are a generic computational framework to define deep learning architectures on graph-structered data. Given a heterogeneous graph $G = (\mathcal{V}, \mathcal{E}, \phi, \psi)$ with initial node embeddings $\{\mathbf{h}_v^{(0)}\}_{v \in \mathcal{V}}$, a single message passing iteration computes updated features $\{\mathbf{h}_v^{(i+1)}\}_{v \in \mathcal{V}}$ from features $\{\mathbf{h}_v^{(i)}\}_{v \in \mathcal{V}}$ given by the previous iteration. One iteration takes the form:

$$\mathbf{h}_v^{(i+1)} = f(\mathbf{h}_v^{(i)}, \{\!\{ g(\mathbf{h}_w^{(i)}) \mid w \in \mathcal{N}(v) \}\!\}), \tag{4}$$

where f and g are arbitrary differentiable functions with optimizable parameters and $\{\!\{\cdot\}\!\}$ an permutation invariant set aggregator, such as mean, max, sum, or a combination. Heterogeneous message passing (Schlichtkrull et al., 2018; Hu et al., 2020) is a *nested* version of Eq. 4, adding an aggregation over all incoming edge types to learn distinct message types:

$$\mathbf{h}_v^{(i+1)} = f_{\phi(v)}\Big(\mathbf{h}_v^{(i)}, \Big\{\!\!\Big\{ f_R(\{\!\{ g_R(\mathbf{h}_w^{(i)}) \mid w \in \mathcal{N}_R(v) \}\!\}) \Big| \forall R = (T, \phi(v)) \in \mathcal{R} \Big\}\!\!\Big\}\Big), \tag{5}$$

where $\mathcal{N}_R(v) = \{w \in \mathcal{V} \mid (w, v) \in \mathcal{E} \text{ and } \psi(w, v) = R\}$ denotes the *R-specific neighborhood* of node $v \in \mathcal{V}$. This formulation supports a wide range of different graph neural network operators, which define the specific form of functions $f_{\phi(v)}, f_R, g_R$ and $\{\!\{\cdot\}\!\}$ (Fey & Lenssen, 2019).

Temporal Message Passing. Given a relational entity graph $G = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{R})$ with attached mapping functions ψ, ϕ, τ and initial node embeddings $\{\mathbf{h}_v^{(0)}\}_{v \in \mathcal{V}}$ and an example specific *seed time* $t \in \mathbb{R}$ (*cf.* Sec. 2.2) , we obtain a set of deep node embeddings $\{\mathbf{h}_v^{(L)}\}_{v \in \mathcal{V}}$ by L consecutive applications of Eq. 5, where we additionally filter R-specific neighborhoods based on their timestamp, *i.e.* replace $\mathcal{N}_R(v)$ with

$$\mathcal{N}_R^{\leq t}(v) = \{w \in \mathcal{V} \mid (w, v) \in \mathcal{E}, \psi(w, v) = R, \text{ and } \tau(w) \leq t\},$$

realized by the temporal sampling procedure presented in Sec. 3.3. The formulation naturally respects time by only aggregating messages from nodes that were available before the given seed time s. The given formulation is agnostic to specific implementations of message passing and supports a wide range of different operators.

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Relational Graph Transformer (RELGT)

- RELGT is a **Transformer architecture** specifically designed to model large-scale, heterogeneous, and temporal relational graphs.
- Each node is tokenized into a structured representation that encodes its attributes, type, temporal context, and structural role.

RELATIONAL GRAPH TRANSFORMER

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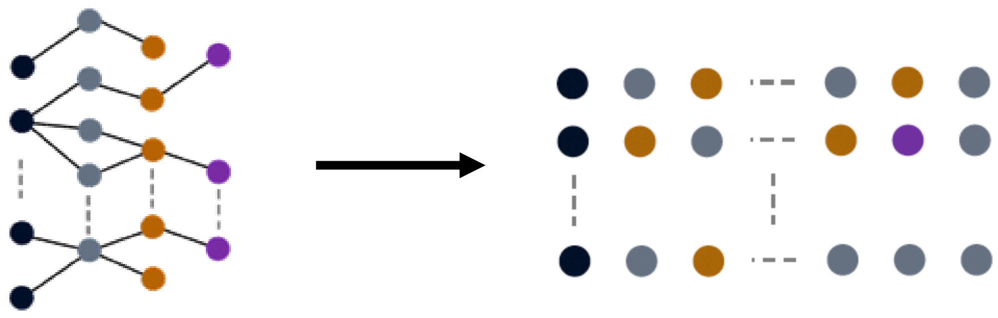
ABSTRACT

Relational Deep Learning (RDL) is a promising approach for building state-of-the-art predictive models on multi-table relational data by representing it as a heterogeneous temporal graph. However, commonly used Graph Neural Network models suffer from fundamental limitations in capturing complex structural patterns and long-range dependencies that are inherent in relational data. While Graph Transformers have emerged as powerful alternatives to GNNs on general graphs, applying them to relational entity graphs presents unique challenges: (i) Traditional positional encodings fail to generalize to massive, heterogeneous graphs; (ii) existing architectures cannot model the temporal dynamics and schema constraints of relational data; (iii) existing tokenization schemes lose critical structural information. Here we introduce the Relational Graph Transformer (RELGT), the first graph transformer architecture designed specifically for relational tables. RELGT employs a novel multi-element tokenization strategy that decomposes each node into five components (features, type, hop distance, time, and local structure), enabling efficient encoding of heterogeneity, temporality, and topology without expensive precomputation. Our architecture combines local attention over sampled subgraphs with global attention to learnable centroids, incorporating both local and database-wide representations. Across 21 tasks from the RelBench benchmark, RELGT consistently matches or outperforms GNN baselines by up to 18%, establishing Graph Transformers as a powerful architecture for Relational Deep Learning[1].



Tokenziation Process of RELGT

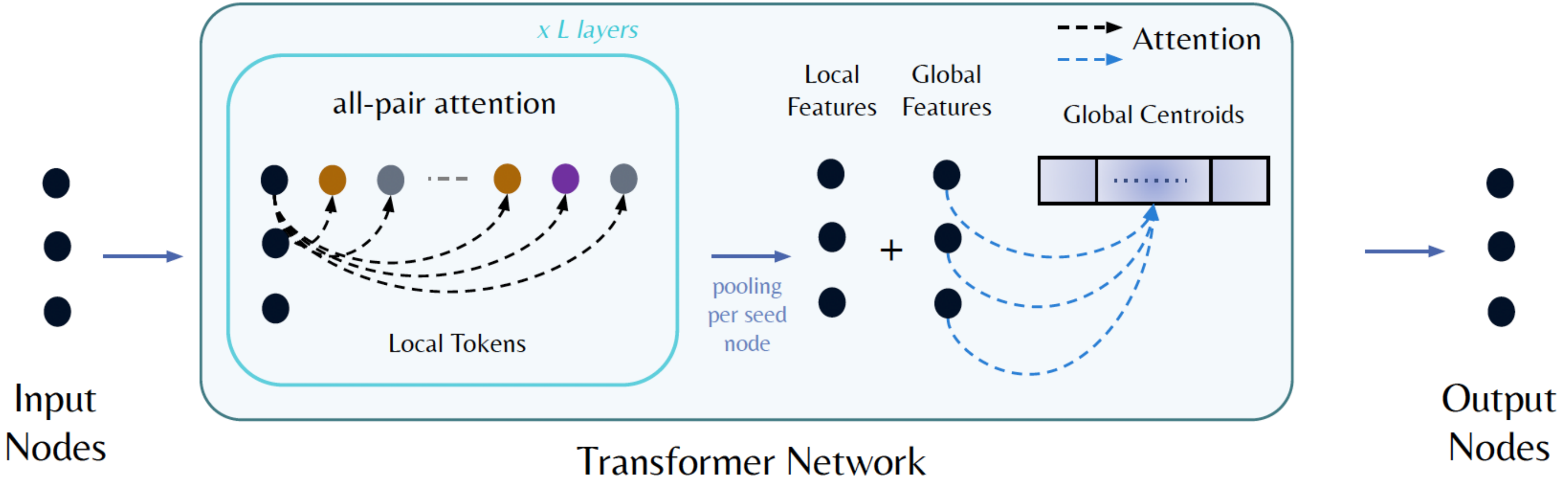
Encoders for Entity Embeddings

- Node Feature Encoder $h_{\text{feat}}(v_j) = \text{MultiModalEncoder}(x_{v_j}) \in \mathbb{R}^d$
- Node Type Encoder $h_{\text{type}}(v_j) = W_{\text{type}} \cdot \text{onehot}(\phi(v_j)) \in \mathbb{R}^d$
- Hop Encoder $h_{\text{hop}}(v_i, v_j) = W_{\text{hop}} \cdot \text{onehot}(p(v_i, v_j)) \in \mathbb{R}^d$
- Time Encoder $h_{\text{time}}(v_i, v_j) = W_{\text{time}} \cdot (\tau(v_j) - \tau(v_i)) \in \mathbb{R}^d$
- Subgraph PE Encoder $h_{\text{pe}}(v_j) = \text{GNN}(A_{\text{local}}, Z_{\text{random}})_j \in \mathbb{R}^d$

$$h_{\text{token}}(v_j) = O \cdot [h_{\text{feat}}(v_j) \,||\, h_{\text{type}}(v_j) \,||\, h_{\text{hop}}(v_i, v_j) \,||\, h_{\text{time}}(v_i, v_j) \,||\, h_{\text{pe}}(v_j)]$$

Relational Graph Transformer

- RELGT includes **local** attention and **global** attention.
- The model is trained end-to-end using suitable **task specific** loss functions.



$$h_{\text{local}}(v_i) = \text{Pool}(\text{FFN}(\text{Attention}(v_i, \{v_j\}_{j=1}^{K}))_L)$$

$$h_{\text{global}}(v_i) = \text{Attention}(v_i, \{c_b\}_{b=1}^{B})$$

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Relational Foundation Model (KumoRFM)

- **KumoRFM**: relational foundation model making predictions over **any relational database** and **any predictive task** without requiring any data- or task-specific training.
- **KumoRFM** is pre-trained based on **RELGT** built on the principles of **RDL**.

KumoRFM: A Foundation Model for In-Context Learning on Relational Data

Corresponding Authors

Matthias Fey* **Vid Kocijan** **Federico Lopez** **Jan Eric Lenssen** **Jure Leskovec***

Relational Foundation Model (KumoRFM)

- KumoRFM is built upon a large-scale **pretrained** Relational Graph Transformer.
- KumoRFM performs **in-context learning** on query-generated relational subgraphs, without retraining the model.
  - It enables prediction on **unseen** relational datasets through in-context learning.

Context Subgraphs /labels
(Train)

$$\tilde{y}_e^{(t)} = \mathrm{KumoRFM}_{\boldsymbol{\theta}}\left(\boxed{\mathcal{G}_k^{\le t}[e]}, \boxed{\left\{\left(\mathcal{G}_k^{\le \hat{t}}[\hat{e}],\ y_{\hat{e}}^{\hat{t}}\right)\right\}_{\substack{\hat{t}<t\\ \hat{e}\sim\mathcal{G}}}}\right),$$

Target Entity
Subgraph (Test)

Predictive Query Language (PQL)

- A Predictive Query is a **declarative syntax** used to define a predictive modeling task in Kumo.
- Surprisingly, it provides interface that interpret NL into PQL (details not provided).

A **Predictive Query** defines a predictive modeling task in Kumo using **PQL (Predictive Query Language)**, a SQL-like syntax that specifies:

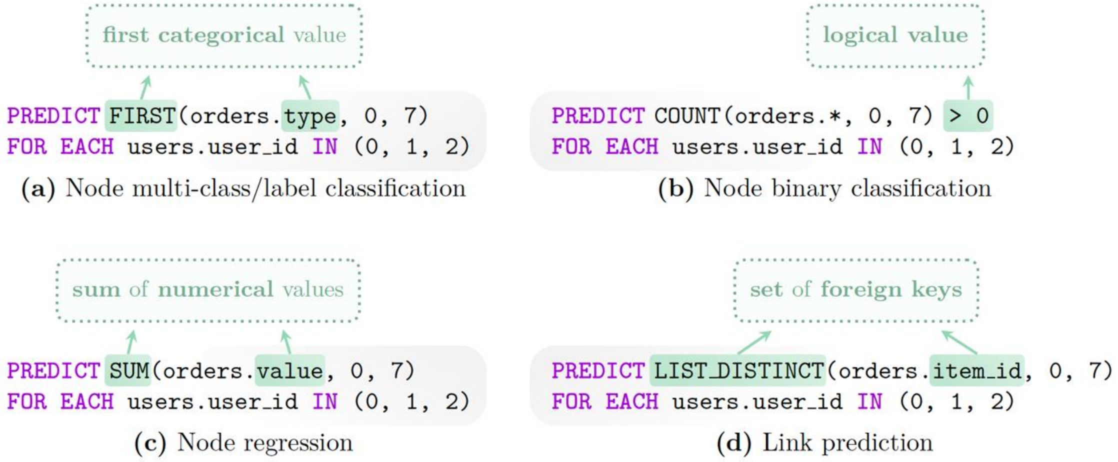
- **Target** – What you want to predict.
- **Entity** – Who you are making predictions for.
- **Filters (optional)** – Constraints on which entities or data to include.

```
PREDICT  SUM(TRANSACTIONS.PRICE, 0, 30)        TARGET
FOR EACH CUSTOMERS.CUSTOMER_ID                 ENTITY
WHERE    COUNT(TRANSACTIONS.*,-30, 0) > 0      FILTER
```

https://kumo.ai/docs/pquery-structure/

Predictive Query Language (PQL)

- Query has **PREDICT, FOR, WHERE clause.**
- PQL **supports various predictive tasks** (regression, binary classification, multi-class classification, and link prediction).



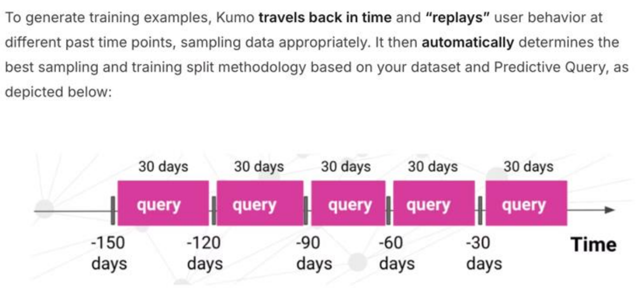
Common Task Types

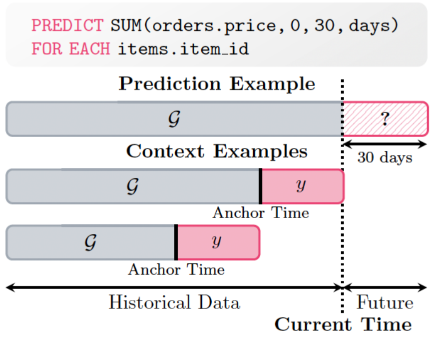
Task Type	Output	PQL Example
Regression	Continuous real number	`PREDICT customers.age FOR EACH customers.customer_id`
Binary Classification	True or False	`PREDICT fraud_reports.is_fraud FOR EACH transactions.id WHERE transactions.type = "bank transfer"`
Multiclass + Multilabel Classification	Class label	`PREDICT FIRST(purchases.type, 0, 7) FOR EACH users.user_id`
Link Prediction	List of items	`PREDICT LIST_DISTINCT(transactions.article_id, 0, 7) RANK TOP 10 FOR EACH customers.customer_id`

Online Context Generation

- k-hop subgraph $g_k^{\le t}[e] \subseteq \mathcal{G}$ is sampled around entity e **up to timestamp t** to be used as input to make a prediction $\tilde{y}_e^t$.
- **Dynamically update** the graph on-the-fly, grouping historical labels into a dynamically constructed training table $\hat{T}$ and attaching it to the entity table.

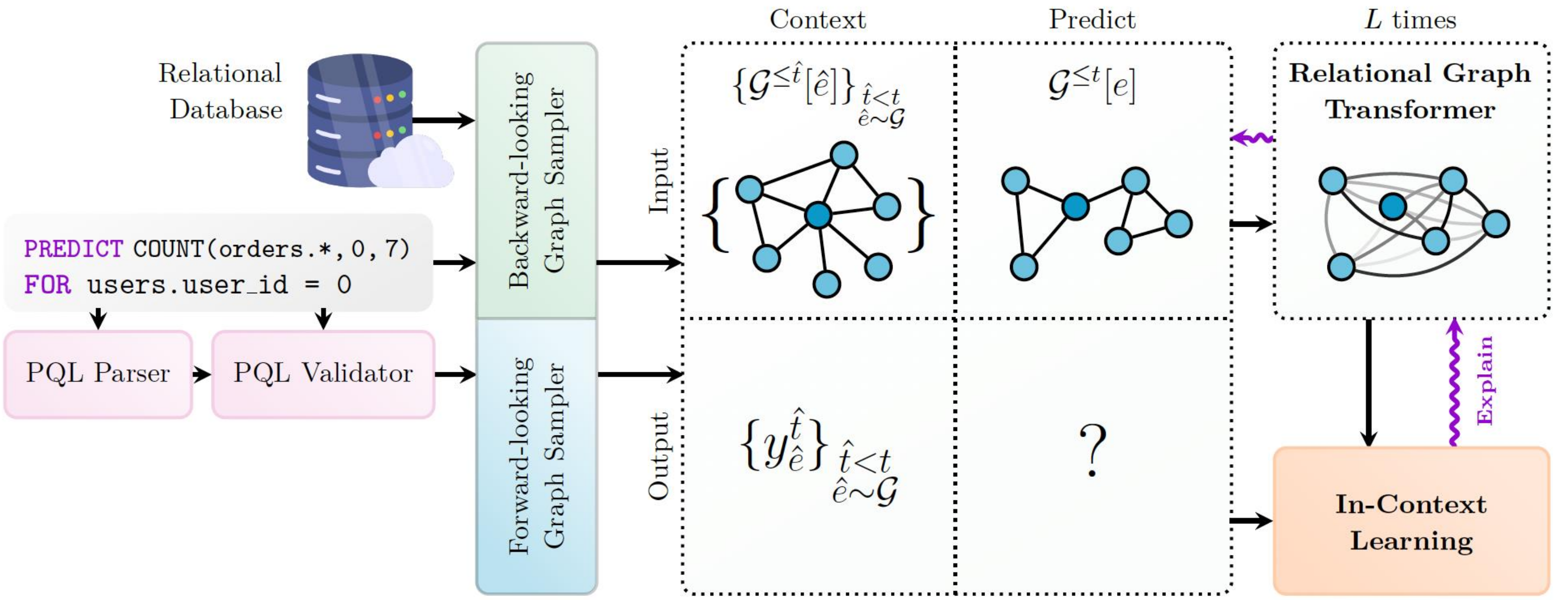
$$\tilde{y}_e^{(t)} = \mathrm{KumoRFM}_{\boldsymbol{\theta}}\left(\mathcal{G}_k^{\le t}[e], \boxed{\left\{\left(\mathcal{G}_k^{\le \hat{t}}[\hat{e}],\ y_{\hat{e}}^{\hat{t}}\right)\right\}_{\substack{\hat{t}<t \\ \hat{e}\sim\mathcal{G}}}}\right),$$





Temporal Sampling Procedures

- **Backward**-looking: to generate the corresponding input subgraphs
- **Forward**-looking: to generate the in-context labels



Relational Foundation Model (KumoRFM-v2)

- Last April, **KumoRFM-v2** was released.
- Slight changes were added for **task-conditioned** relational reasoning.

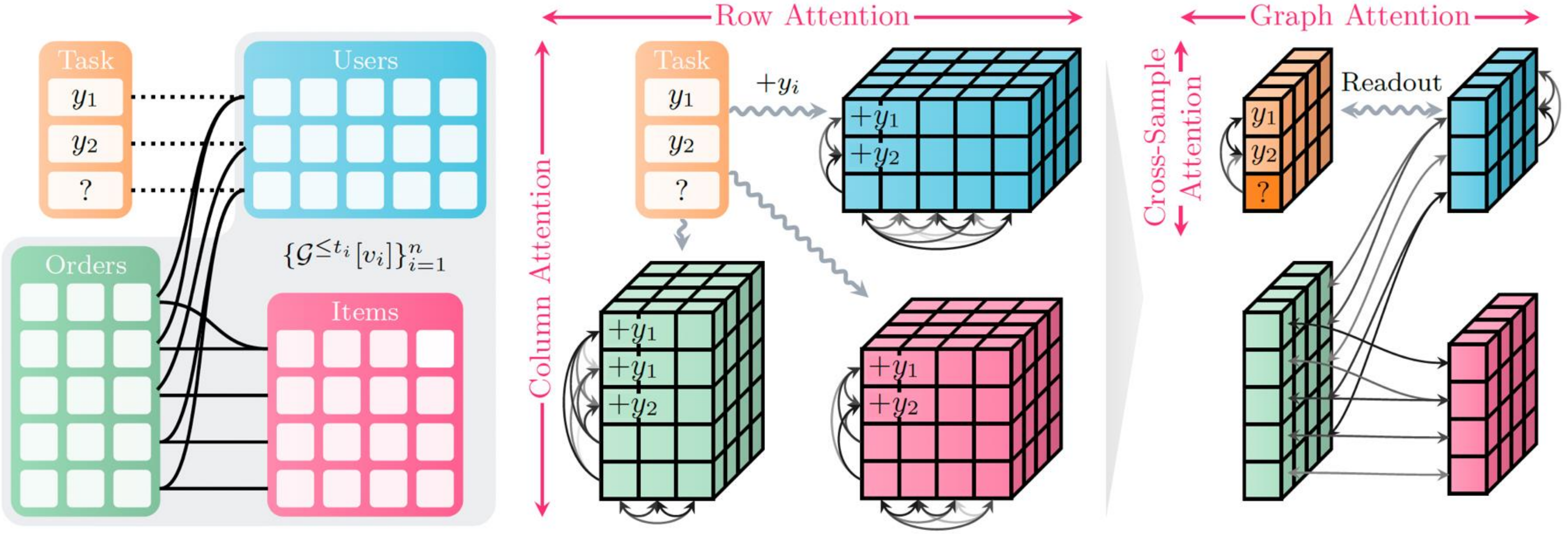
kumo

KumoRFM-2: Scaling Foundation Models for Relational Learning

Valter Hudovernik, Federico López, Vid Kocijan, Akihiro Nitta, Jan Eric Lenssen, Jure Leskovec, Matthias Fey

KumoRFM-2 Architecture

- KumoRFM-2 first computes representations for all rows in the context via **alternating column- and row-wise attention**.
- The representations are enriched via graph attention over **primary-foreign key relationships** and **cross-sample attention** over context examples.



(a) Context Input (b) Reason within Tables (c) Reason across Tables

Key Changes

- Hierarchical Attention Scheme:
  - It introduces early task-conditioned hierarchical attention, enabling **task-aware relational representations** from the row/column feature extraction stage.
- Smarter Context and Lagged Targets:
  - It combines (**local**) self-history and recent **global** database patterns to build more informative task-aware contexts.

Benchmark for KumoRFM-2

- RelBenchV1
- RelBenchV2
- SALT
- 4DBInfer

None of these datasets were used during pre-training, which guarantees **no leakage** of information!

Baselines for KumoRFM-v2

- Supervised Tabular ML
  - Traditional ML
  - AutoML
  - Feature Engineering based
- Naïve
  - Global mean/median
- Supervised Relational ML
  - GNNs / GT
- Foundation Models
  - LLMs
  - Tabular FMs
  - Relational FMs

	Category	Model	Description
SUPERVISED TABULAR ML	*Gradient Boosted Decision Tree*	XGBoost (Chen & Guestrin, 2016)	Ensemble of decision trees trained via gradient boosting, where each tree corrects the residual errors of the previous ones, forming a competitive baseline on tabular data.
		LightGBM (Ke et al., 2017)	
		CatBoost (Prokhorenkova et al., 2018)	
	Deep Tabular ML	CARTE (Kim et al., 2024)	Deep representation learning on tabular data, trained in a (self-)supervised fashion.
	AutoML	AutoGluon (Erickson et al., 2020)	Ensemble of diverse learners (*e.g.*, Neural Networks, GBDTs) with automated hyperparameter optimization. Generally provides best performance among tabular models.
	Manual Feature Engineering	DS+LightGBM	Current gold-standard. A data scientist expert that engineers task-specific features (*e.g.*, temporal aggregations, multi-hop joins) to capture relational dependencies, followed by a tabular model.
		DS+AutoGluon	
		DS+TabPFN-2.5	
		...(Robinson et al., 2024)	
	Automatic Feature Engineering	DFS+XGBoost	Task-agnostic, fixed-function, column-wise encoder to flatten multi-tables based on deep feature synthesis, followed by a tabular model.
		DFS+AutoGluon (Kanter & Veeramachaneni, 2015)	
SUP. RELATIONAL ML	*Graph Neural Network*	GraphSAGE (Hamilton et al., 2017)	Message-passing networks based on different aggregation schemes and information flows, trained in a supervised fashion based on the relational deep learning paradigm.
		GAT (Veličković et al., 2018)	
		PNA (Corso et al., 2020)	
		RelGNN (Chen et al., 2025)	
	Graph Transformer	HGT/HGT_{PE} (Hu et al., 2020)	Self-attention networks, trained in a supervised fashion based on the relational deep learning paradigm.
		RelGT (Dwivedi et al., 2026)	
FOUNDATION MODELS	*Language Model*	LLM_1 (Ranjan et al., 2026)	Language models adapted for relational reasoning through structured prompts and context encoding.
		LLM_2 (Wydmuch et al., 2024)	
	Tabular Foundation Model	TabPFN-2.5 (Hollmann et al., 2025)	In-context learning over single tabular data based on a pre-trained transformer architecture.
		TabICLv2 (Qu et al., 2026)	
	Relational Foundation Model	Griffin (Wang et al., 2025)	Pre-trained relational foundation model that leverages large-scale meta-learning.
		RT_{zero} (Ranjan et al., 2026)	Relational transformer for "zero"-shot predictions, pre-trained via masked token prediction over relational data.
		GNN+TabPFN-2.5 (Meyer et al., 2026)	Randomly initialized GNN to flatten multi-tables, followed by a tabular foundation model.
		RDBLearn (Xu et al., 2026)	Task-agnostic deep feature synthesis, followed by a tabular foundation model.
		KumoRFM-1 (Fey et al., 2025)	The first iteration of KumoRFM.

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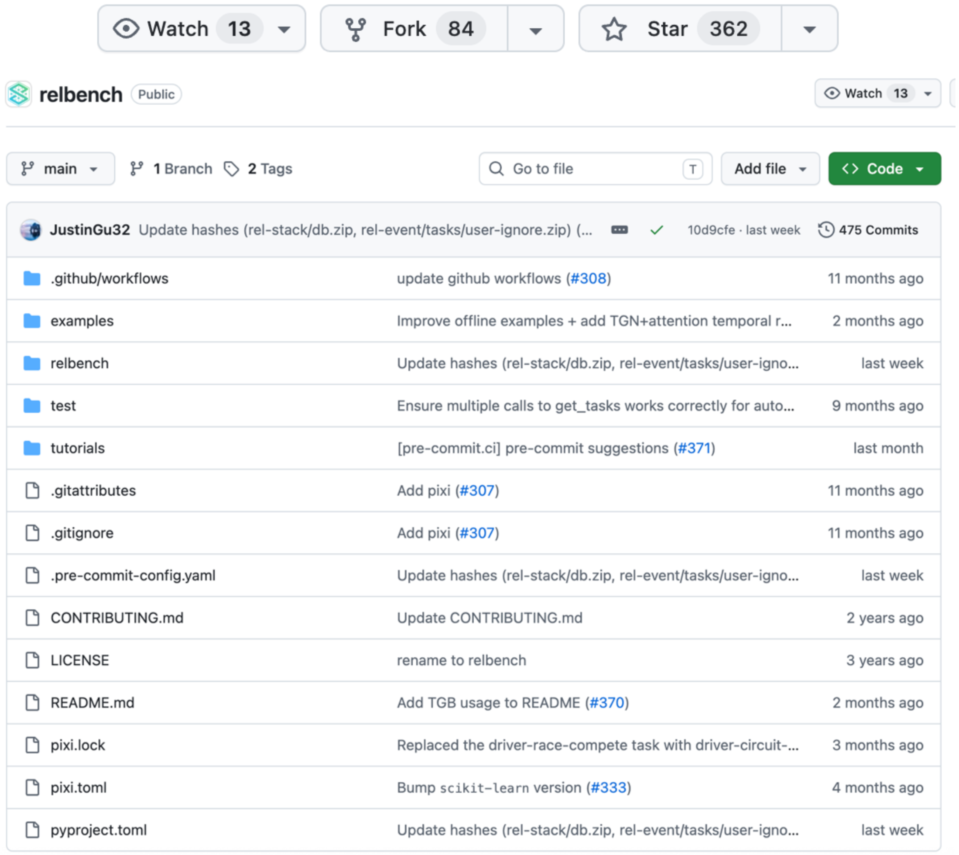
RelBench: Benchmark for Relational Model

- RelBench GitHub page as of May 28, 2026.



RelBench Datasets

- Statistics of RelBench dataset used for KumoRFM.
- RelBench is updated continuously!

v1

Name	Domain	#Tasks	Tables			Timestamp (year-mon-day)		
			#Tables	#Rows	#Cols	Start	Val	Test
rel-amazon	E-commerce	7	3	15,000,713	15	2008-01-01	2015-10-01	2016-01-01
rel-avito	E-commerce	4	8	20,679,117	42	2015-04-25	2015-05-08	2015-05-14
rel-event	Social	3	5	41,328,337	128	1912-01-01	2012-11-21	2012-11-29
rel-f1	Sports	3	9	74,063	67	1950-05-13	2005-01-01	2010-01-01
rel-hm	E-commerce	3	3	16,664,809	37	2019-09-07	2020-09-07	2020-09-14
rel-stack	Social	5	7	4,247,264	52	2009-02-02	2020-10-01	2021-01-01
rel-trial	Medical	5	15	5,434,924	140	2000-01-01	2020-01-01	2021-01-01
Total		30	51	103,466,370	489	/	/	/

v2

Name	Domain	#Tasks	Tables			Timestamp (year-mon-day)		
			#Tables	#Rows	#Cols	Start	Val	Test
rel-salt	Enterprise	8	4	4,257,145	31	2018-01-02	2020-02-01	2020-07-01
rel-arxiv	Academic	4	6	2,146,112	21	2018-01-01	2022-01-01	2023-01-01
rel-ratebeer	Consumer	8	13	13,787,005	221	2000-04-02	2018-09-01	2020-01-01
rel-mimic	Medical	1	6	2,424,751	54	2110-01-14	2168-09-06	2180-10-26
Total		21	29	22,615,013	327	/	/	/

RelBench-v2 Datasets

- RelBench-v2 newly introduces 4 datasets and **autocomplete tasks**.
  - Model learns to fill in a table's missing cells using the surrounding relational schema and historical context.

Dataset	Task name	Task type	#Rows of training table			#Unique Entities	%train/test Entity Overlap	#Dst Entities
			Train	Validation	Test			
rel-amazon	review-rating	auto-reg	11,822,796	806,355	8,217,532	17,255,399	40.5	–
rel-avito	searchstream-click	auto-bcls	2,212,750	1,177,380	924,990	3,976,413	23.8	–
	searchinfo-isuserloggedon	auto-bcls	1,291,566	695,590	592,133	2,579,289	0.0	–
rel-event	event_interest-interested	auto-bcls	14,442	536	420	14,992	94.5	–
	event_interest-not_interested	auto-bcls	14,442	536	420	14,992	94.5	–
	users-birthyear	auto-reg	33,937	1,731	1,002	36,670	0.0	–
rel-f1	results-position	auto-reg	8,997	1,400	4,798	15,195	0.0	–
	qualifying-position	auto-reg	2,228	1,854	5,733	9,815	0.0	–
rel-hm	transactions-price	auto-reg	14,844,291	235,662	266,364	15,346,317	0.0	–
rel-ratebeer	beer_ratings-total_score	auto-reg	10,620,177	1,227,702	2,495,360	14,343,239	0.0	–
rel-salt	item-plant	auto-mcls	1,622,787	293,823	400,206	2,316,816	0.0	–
	item-shippoint	auto-mcls	1,622,787	293,780	398,536	2,315,103	0.0	–
	item-incoterms	auto-mcls	1,622,787	293,891	402,835	2,319,513	0.0	–
	sales-office	auto-mcls	340,491	71,474	88,942	500,907	0.0	–
	sales-group	auto-mcls	340,491	70,224	83,193	493,908	0.0	–
	sales-payterms	auto-mcls	340,491	71,472	88,831	500,794	0.0	–
	sales-shipcond	auto-mcls	340,491	71,398	88,422	500,311	0.0	–
	sales-incoterms	auto-mcls	340,491	71,470	88,925	500,886	0.0	–
rel-stack	badges-class	auto-mcls	448,358	15,105	127,370	590,833	0.0	–
rel-trial	studies-enrollment	auto-reg	233,072	14,470	23,430	270,972	0.0	–
	studies-has_dmc	auto-bcls	202,840	11,983	18,944	233,767	0.0	–
	eligibilities-adult	auto-bcls	234,366	14,470	23,430	272,266	0.0	–
	eligibilities-child	auto-bcls	234,366	14,470	23,430	272,266	0.0	–

Baselines

- Tabular Machine Learning
  - XGBoost, LightGBM
- Naïve method
  - Global mean/median
- GNNs
  - Homogeneous GNNs
  - Heterogeneous GNNs(RGCN, HAN,HGT)
- RDL
  - HeteroGraphSAGE
  - RELGT
- LLM

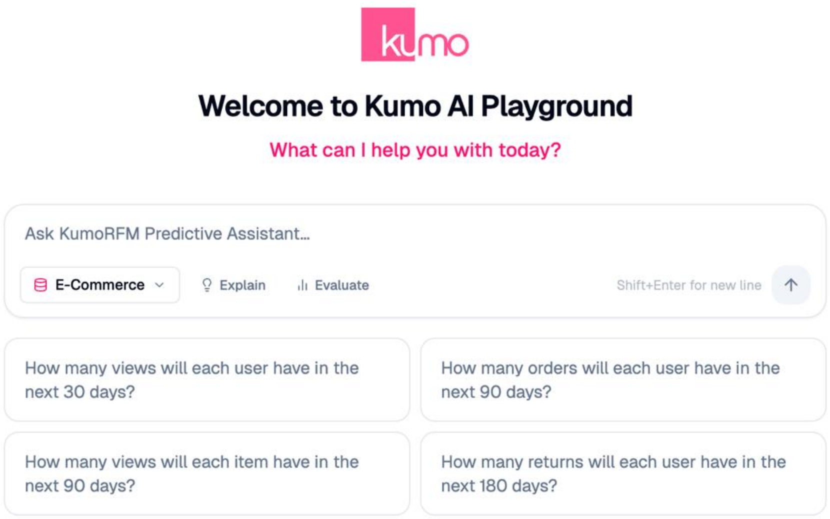
Dataset	Task	Split	Global Zero	Global Mean	Global Median	Entity Mean	Entity Median	LightGBM	RDL	Rel. Gain of RDL
rel-amazon	user-ltv	Val	14.141	20.740	14.141	17.685	15.978	14.141	**12.132**	14.21 %
		Test	16.783	22.121	16.783	19.055	17.423	16.783	**14.313**	14.72 %
	item-ltv	Val	72.096	78.110	59.471	80.466	68.922	55.741	**45.140**	19.02 %
		Test	77.126	81.852	64.234	78.423	66.436	60.569	**50.053**	17.36 %
rel-avito	ad-ctr	Val	0.048	0.048	0.040	0.044	0.044	0.037	**0.037**	2.21 %
		Test	0.052	0.051	0.043	0.046	0.046	**0.041**	0.041	−0.18 %
rel-event	user-attendance	Val	0.262	0.457	0.262	0.296	0.268	0.262	**0.255**	2.65 %
		Test	0.264	0.470	0.264	0.304	0.269	0.264	**0.258**	1.97 %
rel-f1	driver-position	Val	11.083	4.334	4.136	7.181	7.114	3.450	**3.193**	7.44 %
		Test	11.926	4.513	4.399	8.501	8.519	4.170	**4.022**	3.56 %
rel-hm	item-sales	Val	0.086	0.142	0.086	0.117	0.086	0.086	**0.065**	24.50 %
		Test	0.076	0.134	0.076	0.111	0.078	0.076	**0.056**	26.90 %
rel-stack	post-votes	Val	0.062	0.146	0.062	0.102	0.064	0.062	**0.059**	4.19 %
		Test	0.068	0.149	0.068	0.106	0.069	0.068	**0.065**	4.11 %
rel-trial	study-adverse	Val	57.083	75.008	56.786	57.083	57.083	**45.774**	46.290	−1.13 %
		Test	57.930	73.781	57.533	57.930	57.930	**44.011**	44.473	−1.05 %
	site-success	Val	0.475	0.462	0.475	0.447	0.450	0.417	**0.401**	3.87 %
		Test	0.462	0.468	0.462	0.448	0.441	0.425	**0.400**	5.86 %
Average		Val	17.260	19.939	15.051	18.158	16.668	13.330	**11.952**	8.55 %
		Test	18.299	20.393	15.985	18.325	16.801	14.045	**12.631**	8.14 %

Contents

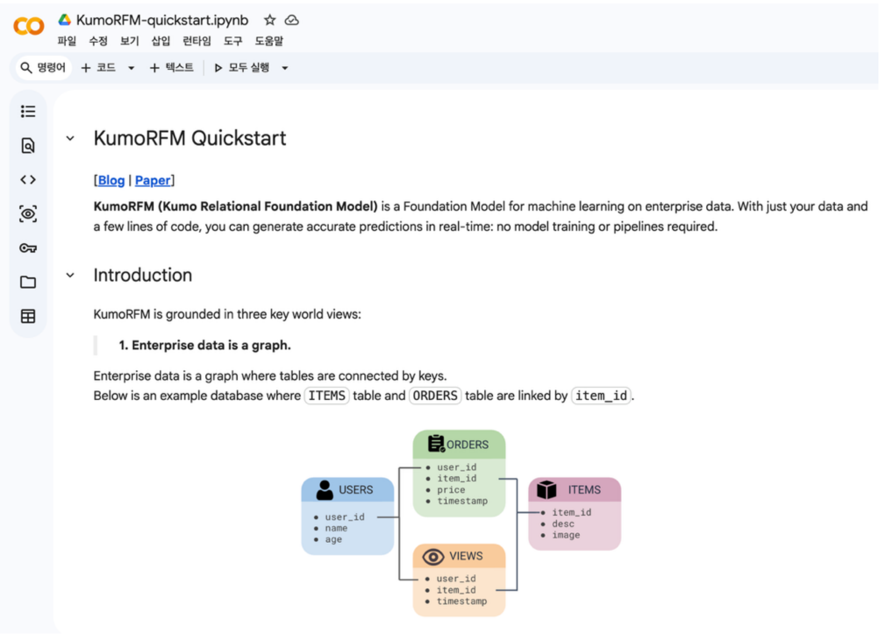
- Introduction to Relational Database
- Relational Deep Learning
- Relational Graph Transformer
- Relational Foundation Model
- Benchmark for RDL
- **Toy example**

Toy Example to Understand KumoRFM

- Let’s play!



[Playground](#) in Kumo.ai



[QuickStart](#) in Colab

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